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DEPARTMENT OF INFORMATION TECHNOLOGY



IT3681 MOBILE APPLICATIONS DEVELOPMENT LABORATORY

Name :

Register Number :

Year & Branch :

Semester :

Academic Year :

ANNAI MIRA COLLEGE OF ENGINEERING AND TECHNOLOGY

NH-46, Chennai-Bengaluru National Highways, Arappakkam,

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CERTIFICATE

This is to certify that the bonafide record of the practical work done by

..... **Register Number** of **III**

year **B.Tech (Information Technology)** submitted for the **APRIL/MAY-2026**

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DEVELOPMENT LABORATORY during the academic year 2025 – 2026.

Staff in -Charge

Head of the Department

Submitted for the practical examination held on -----

INTERNAL EXAMINER

EXTERNAL EXAMINER

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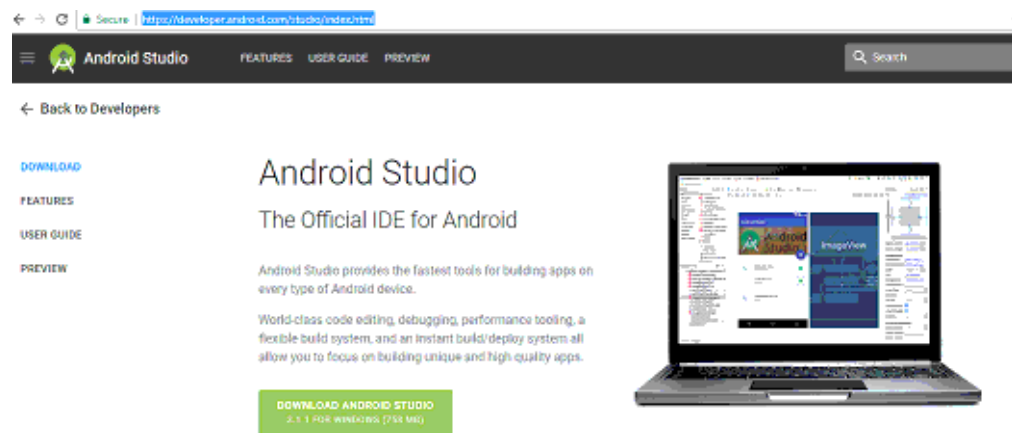
EX. NO: 1

STUDY AND INSTALLATION OF FLUTTER/KOTLIN MULTI- PLATFORM ENVIRONMENT

DATE:

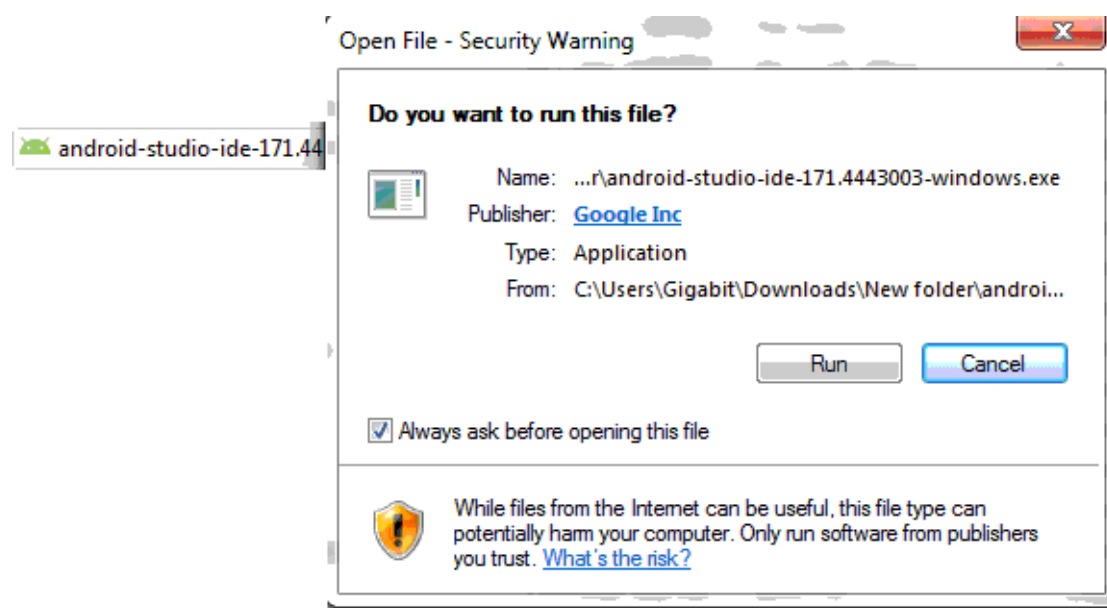
PROCEDURE:

- (i) To develop an android application using Kotlin, we need a Android application development IDE such as Android Studio.
- (ii) Android Studio is an IDE that allows us to develop application at single platform. We can download Android Studio set from Android Studio official link [Download Android Studio](https://developer.android.com/studio/index.html).

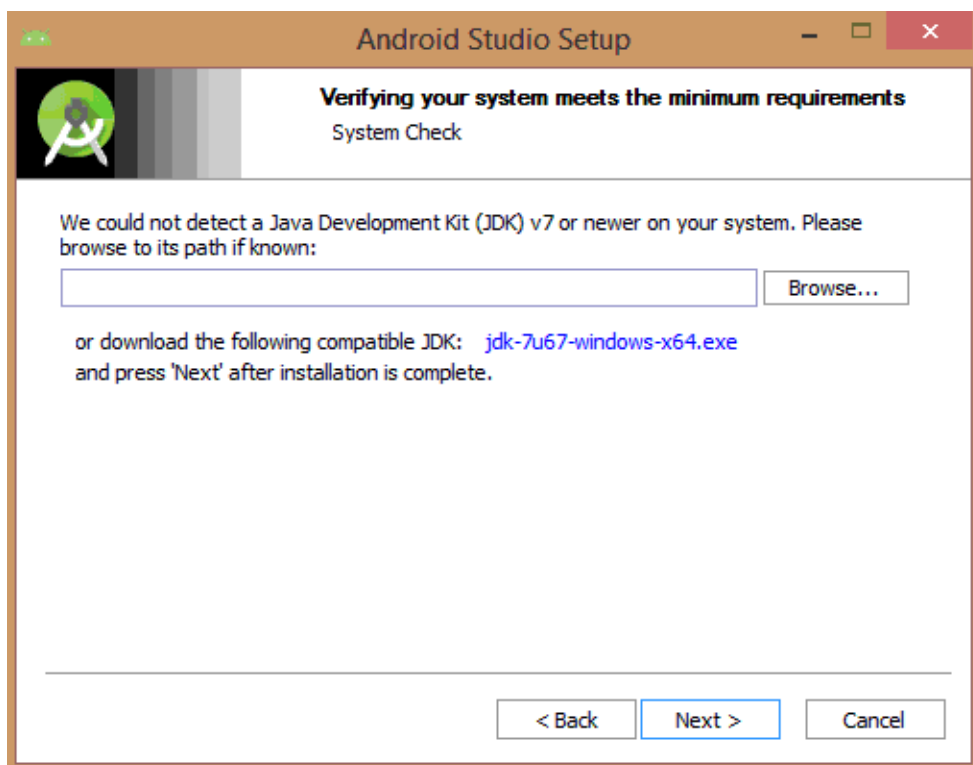
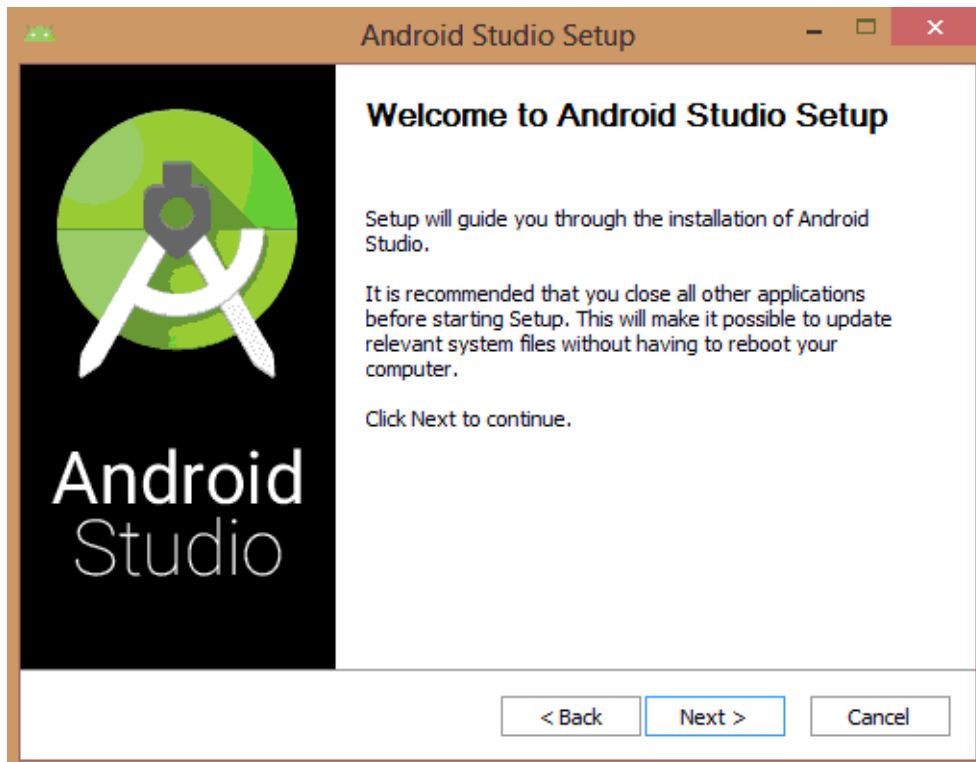


Before installing Android Studio on our machine, we must have Java installed. Look a reference here <https://www.javatpoint.com/how-to-set-path-in-java> to set path of JDK.

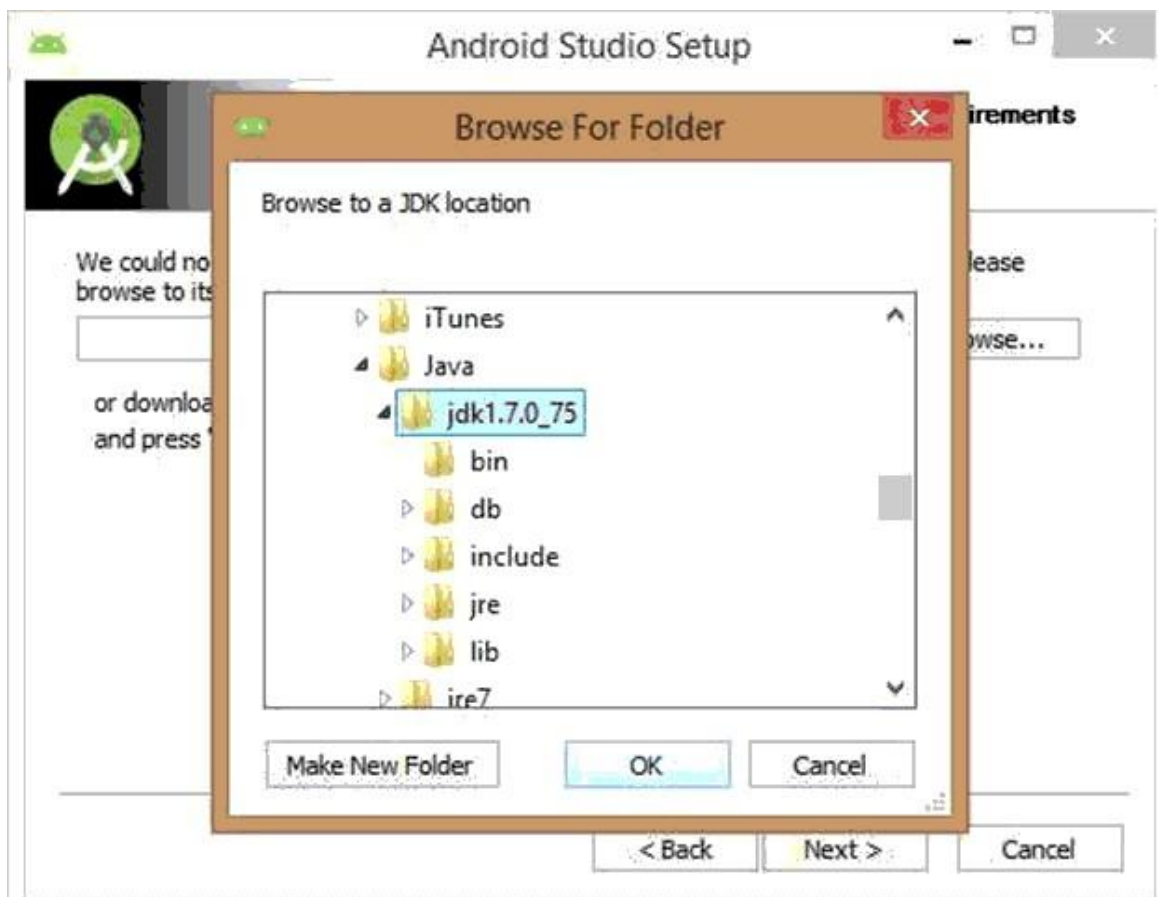
After download Android Studio, run its setup and install it. To install follow the instructions.



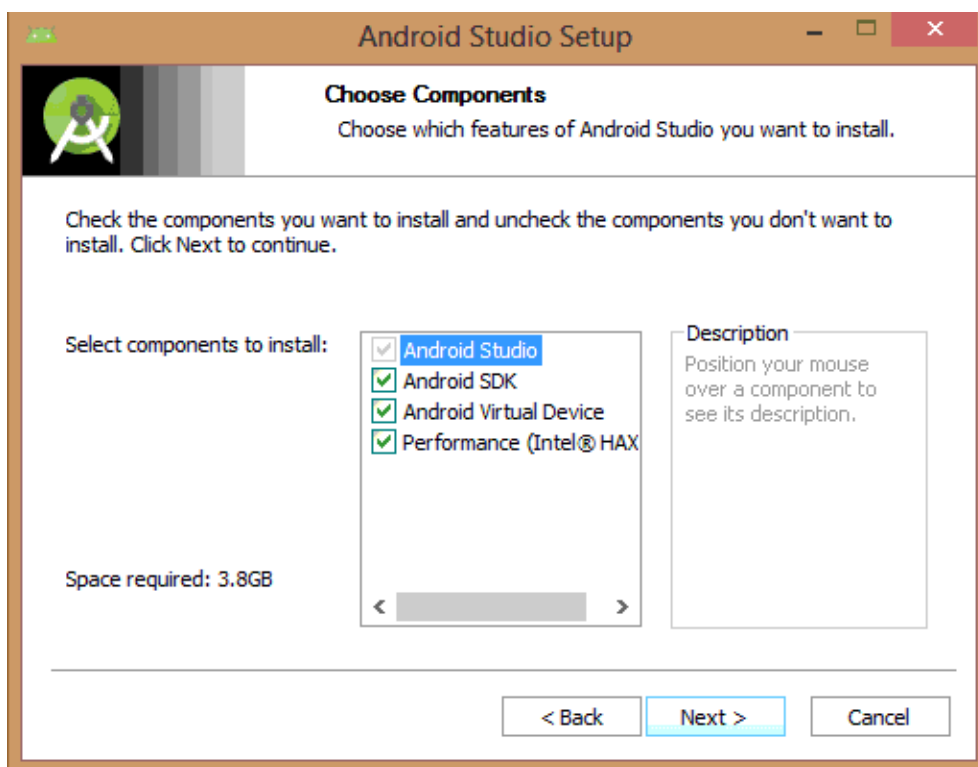
Click Next to proceed.



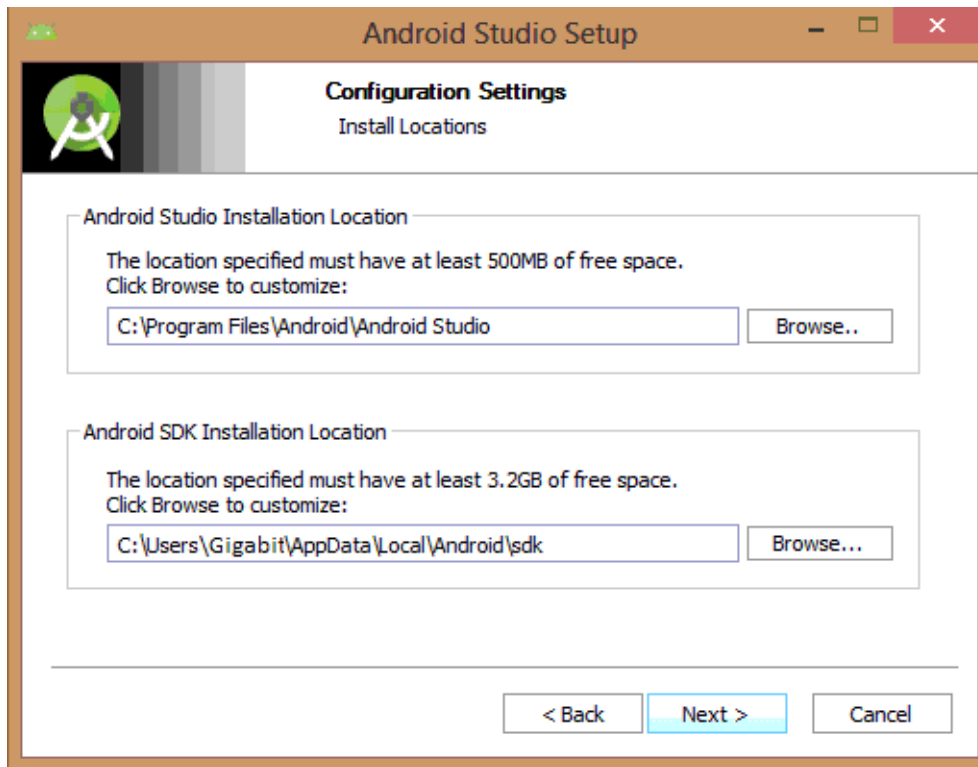
Browse JDK path and click Next



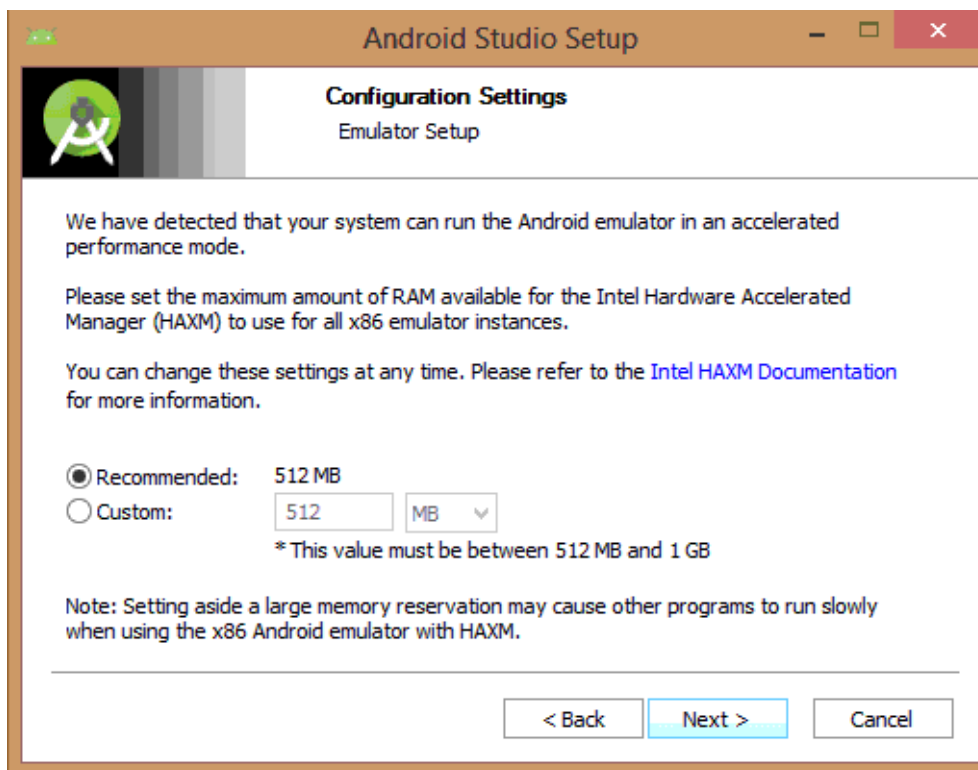
Check the component which we want to install.



Provide Android SDK installation location.

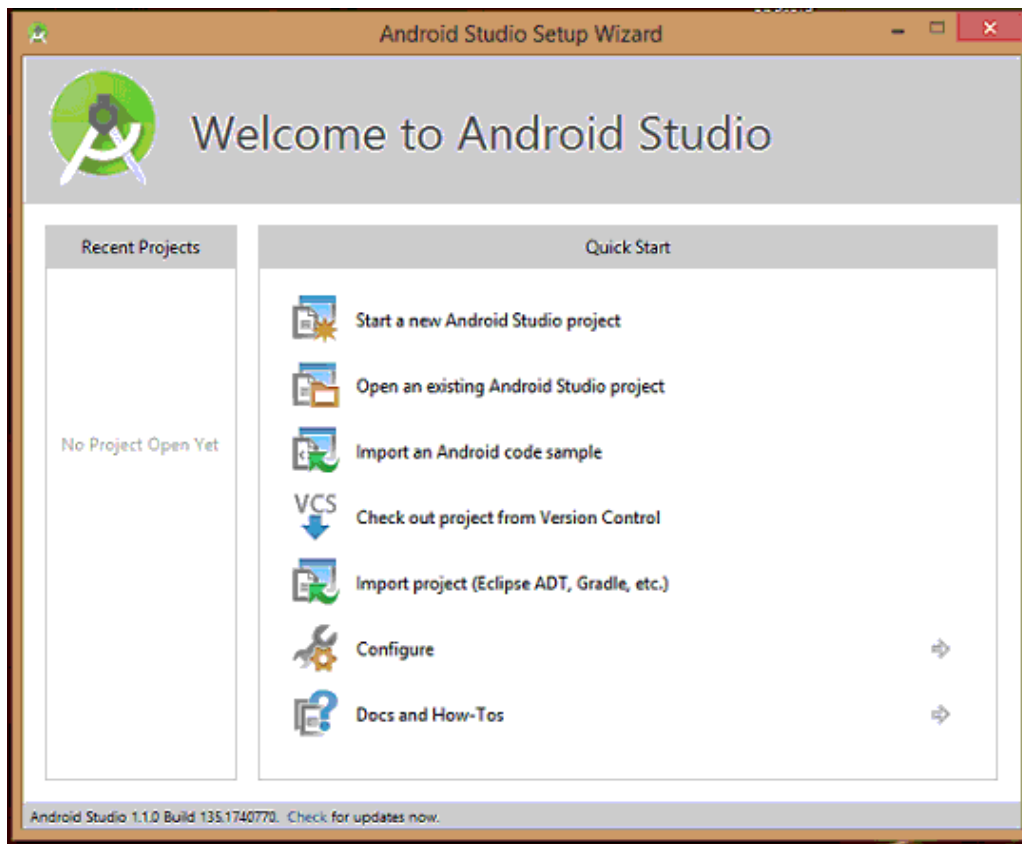


Select the maximum size of RAM to use emulator instances.



Android Studio is successfully installed on our system.

OUTPUT:



RESULT:

Thus the installation of flutter/ kotlin has been successfully installed in multi-platform environment

EX. NO: 2

DEVELOP AN APPLICATION THAT USES WIDGETS, GUI COMPONENTS, FONTS, AND COLORS

DATE:

Java file

```
package com.lab.guicomponents;

import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;

public class MainActivity extends Activity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override

    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }
}
```

XML file

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical"
    android:padding="20dp">
```

```
<EditText  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Edit Text"/>
```

```
<Button  
    android:id="@+id/btn1"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Button"/>
```

```
<CheckBox  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Checkbox"/>
```

```
<RadioButton  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Radio Button"/>
```

```
<TextView  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Green Color"  
    android:textColor="#00ff00"/>
```

```
<TextView  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Red Color"  
    android:textColor="#ff0000"/>
```

```
<TextView  
    android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Blue Color"
```

```
android:textColor="#0000ff"/>
```

```
<TextView
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Size 10"
```

```
android:textColor="#000000"
```

```
android:textSize="10dip"/>
```

```
<TextView
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Size 15"
```

```
android:textColor="#000000"
```

```
android:textSize="15dip"/>
```

```
<TextView
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Size 20"
```

```
android:textColor="#000000"
```

```
android:textSize="20dip"/>
```

```
<TextView
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

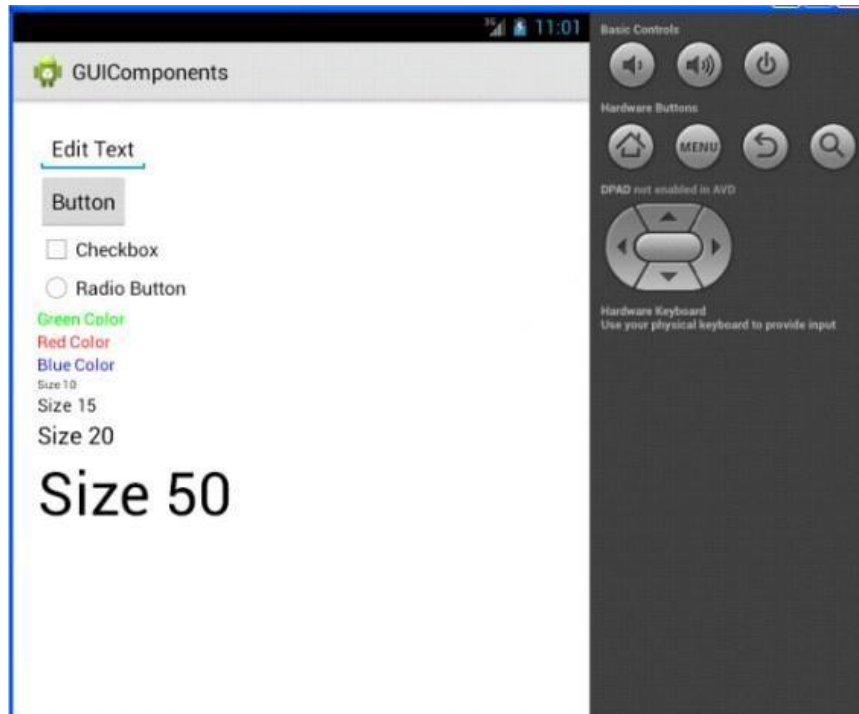
```
android:text="Size 50"
```

```
android:textColor="#000000"
```

```
android:textSize="50dip"/>
```

```
</LinearLayout>
```

OUTPUT:



RESULT:

Thus Developing an application that uses Widgets, GUI components, Fonts, and Colors executed successfully.

EX NO: 3

DEVELOP A NATIVE CALCULATOR APPLICATION

DATE:

JAVA CODE

```
package com.example.rating;

import androidx.appcompat.app.AppCompatActivity;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText firstnum,secondnum;
    TextView r;
    Button bt;
    Button bts;
    Button btm;
    double a,b,c;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        firstnum=(EditText) findViewById(R.id.first);
        secondnum=(EditText) findViewById(R.id.second);
        bt=(Button) findViewById(R.id.buttonadd);
        bts=(Button) findViewById(R.id.buttonsub);
        btm=(Button) findViewById(R.id.buttonmul);
        r=(TextView) findViewById(R.id.result);
```

```
bt.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        a=Double.parseDouble(firstnum.getText().toString());  
        b=Double.parseDouble(secondnum.getText().toString());  
        c=a+b;  
        r.setText("Sum="+c);  
    }  
});  
  
bts.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        a=Double.parseDouble(firstnum.getText().toString());  
        b=Double.parseDouble(secondnum.getText().toString());  
        c=a-b;  
        r.setText("Sub="+c);  
    }  
});  
  
btm.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        a=Double.parseDouble(firstnum.getText().toString());  
        b=Double.parseDouble(secondnum.getText().toString());  
        c=a*b;  
        r.setText("Mul="+c);  
    }  
});  
}
```

XML CODE

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_centerHorizontal="true"
tools:context=".MainActivity">
<TextView
android:id="@+id/textview1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:text="Enter number 1"
android:textSize="18sp" />
<EditText
android:id="@+id/first"
android:text="one"
android:minHeight="48dp"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@id/textview1"
android:layout_alignParentTop="false"
android:layout_alignParentRight="false"
android:layout_centerHorizontal="true"
android:ems="10"
android:inputType="number" />
```

```
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/first"
    android:layout_centerHorizontal="true"
    android:text="Enter Number 2"
    android:textSize="18sp"/>
```

```
<EditText
    android:id="@+id/second"
    android:text="two"
    android:minHeight="48dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView2"
    android:layout_centerHorizontal="true"
    android:ems="10"
    android:inputType="number" />
```

```
<Button
    android:id="@+id/buttonadd"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/second"
    android:layout_centerHorizontal="true"
    android:text="Add"/>
```

```
<Button
    android:id="@+id/buttonsub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

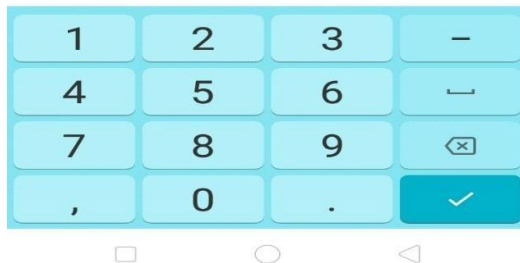
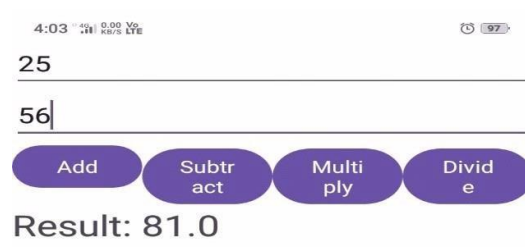
```
    android:layout_below="@id/buttonadd"
    android:layout_centerHorizontal="true"
    android:text="Sub" />

    <Button
        android:id="@+id/buttonmul"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/buttonsub"
        android:layout_centerHorizontal="true"
        android:text="mul" />

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/buttonmul"
        android:layout_centerHorizontal="true"
        android:textSize="24sp" />

</RelativeLayout>
```

OUTPUT:



RESULT:

Thus Develop a native calculator application has been done successfully.

EX NO: 4

DEVELOP A GAMING APPLICATION THAT USES 2-D ANIMATIONS AND GESTURES

DATE:

JAVA CODE

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.ImageView;

public class MainActivity extends AppCompatActivity {
    ImageView imageView;
    Button blinkBTN, rotateBTN, fadeBTN, moveBTN, slideBTN, zoomBTN,
    stopBTN;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        imageView = findViewById(R.id.imageview);
        blinkBTN = findViewById(R.id.BTNblink);
        rotateBTN = findViewById(R.id.BTNrotate);
        fadeBTN = findViewById(R.id.BTNfade);
        moveBTN = findViewById(R.id.BTNmove);
        slideBTN = findViewById(R.id.BTNslide);
        zoomBTN = findViewById(R.id.BTNzoom);
        stopBTN = findViewById(R.id.BTNstop);

        blinkBTN.setOnClickListener(new View.OnClickListener() {
```

```

        @Override
        public void onClick(View v) {
            // To add blink animation

            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.blink_animation);
            imageView.startAnimation(animation);
        }
    });

    rotateBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add rotate animation

            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.rotate_animation);
            imageView.startAnimation(animation);
        }
    });

    fadeBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add fade animation

            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.fade_animation);
            imageView.startAnimation(animation);
        }
    });

    moveBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add move animation

```

```

        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.move_animation);
        imageView.startAnimation(animation);
    }
});
slideBTN.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        // To add slide animation
        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.slide_animation);
        imageView.startAnimation(animation);
    }
});
zoomBTN.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.zoom_animation);
        imageView.startAnimation(animation);
    }
});
stopBTN.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        imageView.clearAnimation();
    }
});
}
}

```

XML CODE

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView

        android:id="@+id/imageview"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"
        android:contentDescription="@string/app_name"
        android:src="@drawable/gfgimage" />

    <LinearLayout

        android:id="@+id/linear1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/imageview"
        android:layout_marginTop="30dp"
        android:orientation="horizontal"
        android:weightSum="3">

        <!--To start the blink animation of the image-->

        <Button

            android:id="@+id/BTNblink"
            style="@style/TextAppearance.AppCompat.Widget.Button"
            android:layout_width="0dp"
```

```
android:layout_height="wrap_content"
```

```
android:layout_margin="10dp"
```

```
android:layout_weight="1"
```

```
android:padding="3dp"
```

```
android:text="@string/blink"
```

```
android:textColor="@color/white" />
```

```
<!--To start the rotate animation of the image-->
```

```
<Button
```

```
android:id="@+id/BTNrotate"
```

```
style="@style/TextAppearance.AppCompat.Widget.Button"
```

```
android:layout_width="0dp"
```

```
android:layout_height="wrap_content"
```

```
android:layout_margin="10dp"
```

```
android:layout_weight="1"
```

```
android:padding="3dp"
```

```
android:text="@string/clockwise"
```

```
android:textColor="@color/white" />
```

```
<!--To start the fading animation of the image-->
```

```
<Button
```

```
android:id="@+id/BTNfade"
```

```
style="@style/TextAppearance.AppCompat.Widget.Button"
```

```
android:layout_width="0dp"
```

```
android:layout_height="wrap_content"
```

```
android:layout_margin="10dp"
```

```
android:layout_weight="1"
```

```
android:padding="3dp"
```

```
android:text="@string/fade"
```

```
android:textColor="@color/white" />
```

```
</LinearLayout>
```

<LinearLayout

android:id="@+id/linear2"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:layout_below="@id/linear1"

android:layout_marginTop="30dp"

android:orientation="horizontal"

android:weightSum="3">

<!--To start the move animation of the image-->

<Button

android:id="@+id/BTNmove"

style="@style/TextAppearance.AppCompat.Widget.Button"

android:layout_width="0dp"

android:layout_height="wrap_content"

android:layout_margin="10dp"

android:layout_weight="1"

android:padding="3dp"

android:text="@string/move"

android:textColor="@color/white" />

<!--To start the slide animation of the image-->

<Button

android:id="@+id/BTNslide"

style="@style/TextAppearance.AppCompat.Widget.Button"

android:layout_width="0dp"

android:layout_height="wrap_content"

android:layout_margin="10dp"

android:layout_weight="1"

android:padding="3dp"

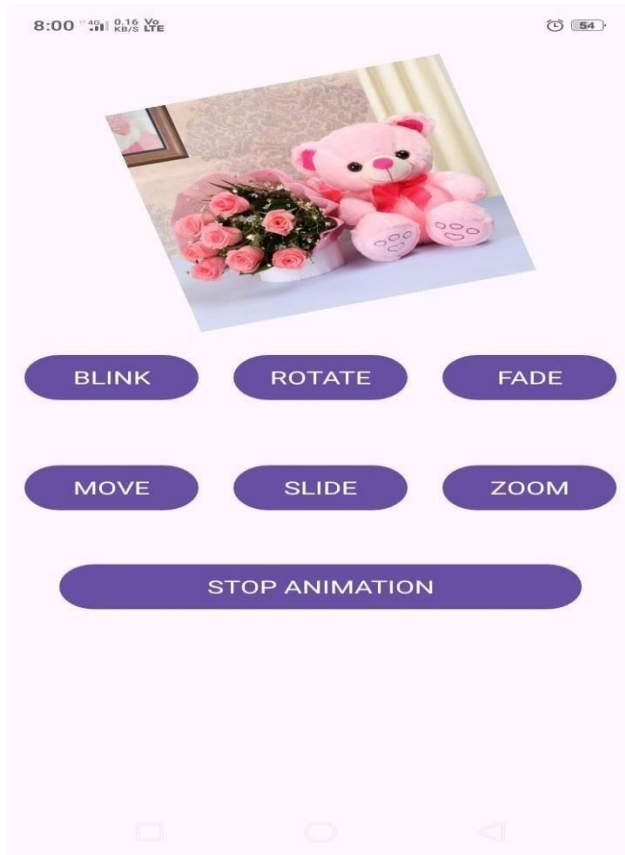
android:text="@string/slide"

```

        android:textColor="@color/white" />
<!--To start the zoom animation of the image-->
<Button
    android:id="@+id/BTNzoom"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/zoom"
    android:textColor="@color/white" />
</LinearLayout>
<!--To stop the animation of the image-->
<Button
    android:id="@+id/BTNstop"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/linear2"
    android:layout_marginLeft="30dp"
    android:layout_marginTop="30dp"
    android:layout_marginRight="30dp"
    android:text="@string/stop_animation" />
</RelativeLayout>

```

OUTPUT:



RESULT:

Thus Developing a gaming application that uses 2-D animations and gestures has been developed completely

EX NO: 5

DEVELOP A MOVIE RATING APPLICATION (SIMILAR TO IMDB)

DATE:

JAVA CODE

```
package com.example.ra;

import android.graphics.Color;
import android.graphics.PorterDuff;
import android.graphics.drawable.LayerDrawable;
import androidx.appcompat.app.AppCompatActivity;
//import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.RatingBar;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    RatingBar rt;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //binding MainActivity.java with activity_main.xml file
        rt = (RatingBar) findViewById(R.id.ratingBar);

        //finding the specific RatingBar with its unique ID
        LayerDrawable stars=(LayerDrawable)rt.getProgressDrawable();

        //Use for changing the color of RatingBar
        stars.getDrawable(2).setColorFilter(Color.YELLOW,
        PorterDuff.Mode.SRC_ATOP);
    }

    public void Call(View v)
```

```

{
// This function is called when button is clicked.

// Display ratings, which is required to be converted into string
first.

TextView t = (TextView)findViewById(R.id.textView2);
t.setText("You Rated :"+String.valueOf(rt.getRating()));
}
}

```

XML CODE

```

<?xml version="1.0" encoding="utf-8"?>

<androidx.coordinatorlayout.widget.CoordinatorLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:fitsSystemWindows="true"
tools:context=".MainActivity">

<RatingBar
android:id="@+id/ratingBar"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="104dp"
android:background="#00AA00"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toTopOf="parent"
tools:layout_constraintLeft_creator="1"

```

```
tools:layout_constraintRight_creator="1"
tools:layout_constraintTop_creator="1" />
<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Rate Me!!!"
    android:textColor="@android:color/background_dark"
    android:textSize="30sp"
    android:textStyle="bold|italic"
    tools:layout_editor_absoluteX="127dp"
    tools:layout_editor_absoluteY="28dp" />
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="148dp"
    android:textColorHint="#FF0000"
    android:textSize="24sp"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/ratingBar"
    tools:layout_constraintRight_creator="1"
    tools:layout_constraintLeft_creator="1" />
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="50dp"
```

```
android:layout_marginTop="50dp"
android:background="@android:color/holo_red_dark"
android:onClick="Call"
android:text="Submit"
android:textColor="@android:color/background_light"
android:textStyle="bold|italic"
app:layout_constraintBottom_toTopOf="@+id/textView2"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toBottomOf="@+id/ratingBar"
tools:layout_constraintBottom_creator="1"
tools:layout_constraintLeft_creator="1"
tools:layout_constraintRight_creator="1"
tools:layout_constraintTop_creator="1" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

OUTPUT:

OUTPUT



RESULT:

Thus Develop a movie rating application (similar to IMDB) completely successfully.

EX NO: 6

DEVELOP AN APPLICATION TO CONNECT TO A WEB SERVICE

DATE:

AND TO RETRIEVE DATA WITH HTTP

JAVA CODE:

```
package com.example.webv;

import android.os.Bundle;
import android.webkit.WebView;
import android.webkit.WebViewClient;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Find the WebView by its unique ID
        WebView webView = findViewById(R.id.web);
        // loading https://www.geeksforgeeks.org url in the WebView.
        webView.loadUrl("https://www.geeksforgeeks.org");
        // this will enable the javascript.
        webView.getSettings().setJavaScriptEnabled(true);
        // WebViewClient allows you to handle
        // onPageFinished and override Url loading.
        webView.setWebViewClient(new WebViewClient());
    }
}
```

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout

xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context="com.gfg.android.examplekotlin.MainActivity">
<!-- unique ID of WebView -- >
<WebView
android:id="@+id/web"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:layout_editor_absoluteX="8dp"
tools:layout_editor_absoluteY="8dp" />
</RelativeLayout>
```

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET" />
```

OUTPUT:



RESULT:

Thus Develop an application to connect to a web service and to retrieve data with HTTP has been executed successfully.

EX NO: 7

DEVELOP A SIMPLE SHOPPING APPLICATION

DATE:

JAVA CODE:

```
package com.example.grocerylist.Adapter;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.recyclerview.widget.RecyclerView;
import com.example.grocerylist.Database.Entity.GroceryItems;
import com.example.grocerylist.R;
import com.example.grocerylist.UI.GroceryViewModel;
import com.example.grocerylist.databinding.GroceryadapterBinding;

public class GroceryAdapter extends
RecyclerView.Adapter<GroceryAdapter.GroceryViewHolder> {

    private List<GroceryItems> list;
    private GroceryViewModel viewModel;

    public GroceryAdapter(List<GroceryItems> list, GroceryViewModel viewModel) {
        this.list = list;
        this.viewModel = viewModel;
    }

    @Override
    public GroceryViewHolder onCreateViewHolder(ViewGroup parent, int viewType)
    {
        GroceryadapterBinding binding =
GroceryadapterBinding.inflate(LayoutInflater.from(parent.getContext()), parent,
false);
```

```
        return new GroceryViewHolder(binding);
    }
```

```
    @Override
    public int getItemCount() {
        return list.size();
    }
```

```
    @Override
    public void onBindViewHolder(GroceryViewHolder holder, int position) {
        GroceryItems currentPosition = list.get(position);
        holder.binding.txtItemName.setText(currentPosition.getItemName());

        holder.binding.txtItemPrice.setText(String.valueOf(currentPosition.getItemPrice()));

        holder.binding.txtItemQuantity.setText(String.valueOf(currentPosition.getItemQuantity()));

        holder.binding.ibDelete.setOnClickListener(v ->
            viewModel.delete(currentPosition));

        if (position == list.size() - 1) {
            int totalCost = 0;
            for (int i = 0; i < list.size(); i++) {
                totalCost += list.get(i).getItemPrice();
            }
            holder.binding.txtItemTotalCost.setVisibility(View.VISIBLE);
            holder.binding.txtTotalCostTitle.setVisibility(View.VISIBLE);
            holder.binding.txtItemTotalCost.setText(String.valueOf(totalCost));
        }
    }
}
```

```

static class GroceryViewHolder extends RecyclerView.ViewHolder {
    private final GroceryadapterBinding binding;

    public GroceryViewHolder(GroceryadapterBinding binding) {
        super(binding.getRoot());
        this.binding = binding;
    }
}
}

```

XML CODE:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="125dp"
    android:background="@drawable/adapter1">

    <!-- To display item name in recycler view -->
    <TextView
        android:id="@+id/txtItemName"
        android:layout_width="0dp"
        android:layout_height="53dp"
        android:layout_marginTop="16dp"
        android:layout_marginEnd="15dp"
        android:layout_marginRight="15dp"
        android:layout_marginBottom="17dp"

```

```
android:fontFamily="@font/rokkitt"
android:text="@string/itemName"
android:textColor="@color/white"
android:textSize="35sp"
app:layout_constraintBottom_toTopOf="@+id/txtTotalCostTitle"
app:layout_constraintEnd_toStartOf="@+id/txtItemQuantity"
app:layout_constraintStart_toEndOf="@+id/cbItemCheck"
app:layout_constraintTop_toTopOf="parent" />
```

<!-- To display item quantity -->

<TextView

```
android:id="@+id/txtItemQuantity"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginEnd="34dp"
android:layout_marginRight="34dp"
android:layout_marginBottom="9dp"
android:text="@string/itemQuantity"
android:textColor="@color/white"
android:textSize="25sp"
app:layout_constraintBottom_toBottomOf="@+id/txtItemName"
app:layout_constraintEnd_toStartOf="@+id/txtItemPrice"
app:layout_constraintStart_toEndOf="@+id/txtItemName"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="1.0" />
```

<!-- To display item price -->

<TextView

```
android:id="@+id/txtItemPrice"
```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="26dp"
android:layout_marginEnd="22dp"
android:layout_marginRight="22dp"
android:layout_marginBottom="26dp"
android:text="@string/itemPrice"
android:textColor="@color/white"
android:textSize="25sp"
android:textStyle="bold"
app:layout_constraintBottom_toTopOf="@+id/txtItemTotalCost"
app:layout_constraintEnd_toStartOf="@+id/ibDelete"
app:layout_constraintStart_toEndOf="@+id/txtItemQuantity"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.0" />
```

<CheckBox

```
android:id="@+id/cbItemCheck"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="29dp"
android:layout_marginLeft="29dp"
android:layout_marginEnd="16dp"
android:layout_marginRight="16dp"
android:background="@color/white"
android:shadowColor="@color/black"
app:layout_constraintBaseline_toBaselineOf="@+id/txtItemName"
app:layout_constraintEnd_toStartOf="@+id/txtItemName"
app:layout_constraintStart_toStartOf="parent" />
```

<!-- This button is used to delete grocery item -->

<ImageButton

```
    android:id="@+id/ibDelete"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="26dp"
    android:layout_marginRight="26dp"
    android:background="@color/black"
    android:src="@drawable/ic_action_delete"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toEndOf="@+id/txtItemPrice"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.257" />
```

<!-- To display total cost of grocery items -->

<TextView

```
    android:id="@+id/txtItemTotalCost"
    android:layout_width="100dp"
    android:layout_height="60dp"
    android:background="@drawable/adapter2"
    android:padding="8dp"
    android:paddingLeft="12dp"
    android:text="@string/totalCost"
    android:textSize="30sp"
    android:textStyle="bold"
    android:visibility="invisible"
    app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toEndOf="@+id/txtTotalCostTitle"
app:layout_constraintTop_toTopOf="@+id/txtTotalCostTitle"
app:layout_constraintVertical_bias="0.875" />
```

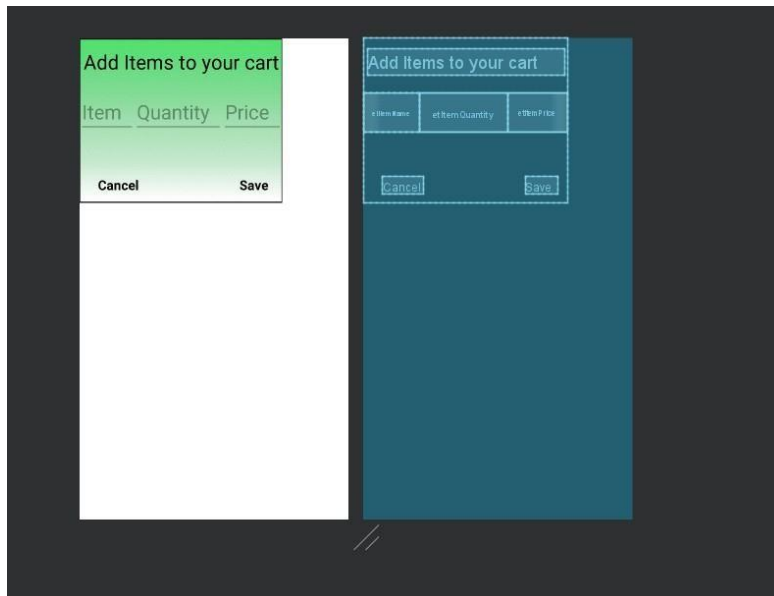
<!-- This text view is used to add statement for total cost -->

<TextView

```
    android:id="@+id/txtTotalCostTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="77dp"
    android:layout_marginLeft="77dp"
    android:layout_marginEnd="149dp"
    android:layout_marginRight="149dp"
    android:layout_marginBottom="16dp"
    android:text="@string/totalCostTitle"
    android:textColor="@color/white"
    android:textSize="20dp"
    android:visibility="invisible"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toStartOf="@+id/txtItemTotalCost"
    app:layout_constraintStart_toStartOf="parent" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

OUTPUT:



RESULT:

Thus Develop a simple shopping application successfully developed.

EX NO: 8

DESIGN A WEB SERVER SUPPORTING PUSH NOTIFICATIONS

DATE:

JAVA CODE:

```
package com.example.notification;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import android.app.NotificationManager;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        button=findViewById(R.id.click);
        button.setOnClickListener(new View.OnClickListener() {

            @Override
            public void onClick(View v) {

                NotificationCompat.Builder
                mbuilder=(NotificationCompat.Builder)
                new NotificationCompat.Builder(getApplicationContext())
                .setSmallIcon(R.drawable.one)
                .setContentTitle("Notification")
                .setContentText("this is noti");

                NotificationManager notificationManager=(NotificationManager)
                getSystemService(NOTIFICATION_SERVICE);
```

```
notificationManager.notify(0,mbuilder.build());
```

```
}
```

```
});
```

```
}
```

```
}
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<Button
```

```
android:id="@+id/click"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

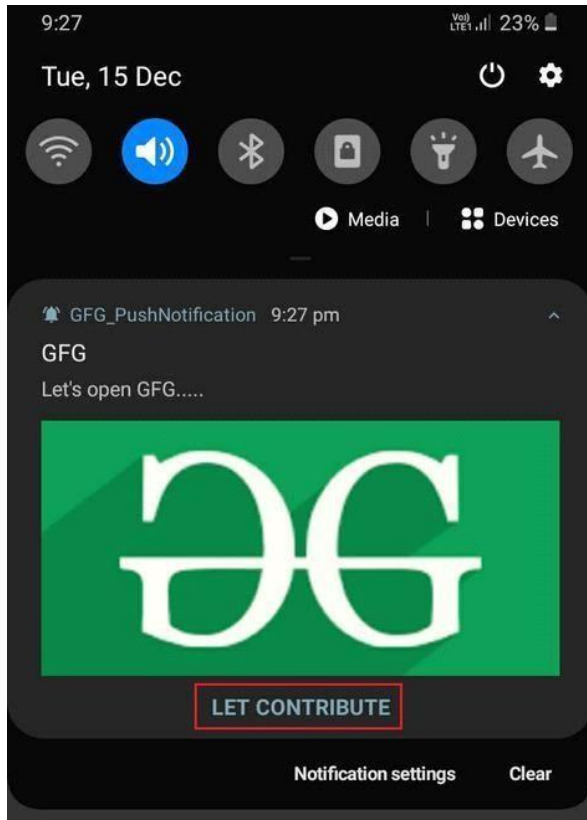
```
android:contentDescription="button"
```

```
android:layout_centerInParent="true"
```

```
android:text="notification"
```

```
</RelativeLayout>
```

OUTPUT:



RESULT:

Thus Design a web server supporting push notifications has been executed successfully.

EX NO: 9

DEVELOP AN APPLICATION BY INTEGRATING GOOGLE MAPS

DATE:

JAVA CODE:

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;

public class MainActivity extends AppCompatActivity implements
OnMapReadyCallback {
    private GoogleMap mMap;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        SupportMapFragment mapFragment = (SupportMapFragment)
getSupportFragmentManager().findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }
    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        // Add a marker in Sydney and move the camera
        LatLng mayiladuthurai = new LatLng(11.1035, 79.6521);
        mMap.addMarker(new MarkerOptions().position(mayiladuthurai).title("Marker in
Mayiladuthurai"));
        mMap.moveCamera(CameraUpdateFactory.newLatLng(mayiladuthurai));

    }
}
```

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <fragment
        android:id="@+id/map"
        android:name="com.google.android.gms.maps.SupportMapFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</RelativeLayout>
```

ANDROID MANIFEST CODE:

The below code must be in application tag

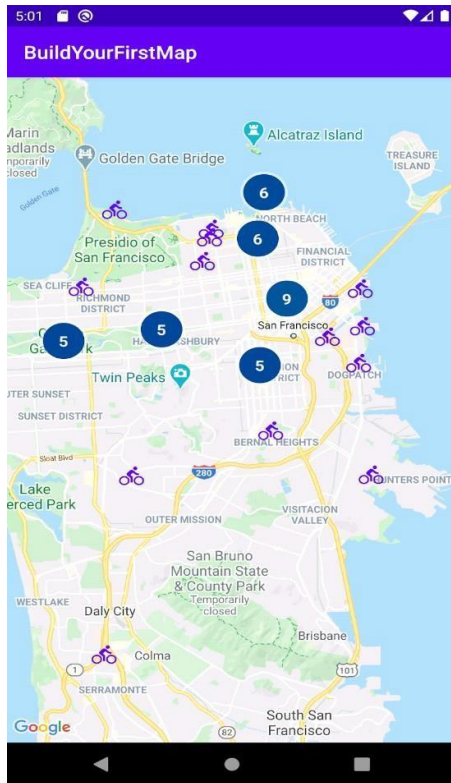
```
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value="// your API "/>
```

Build.gradle.kts(Module:app)

The code must be in dependencies tag

```
implementation("com.google.android.gms:play-services-maps:18.1.0")
```

OUTPUT:



RESULT:

Thus Developing an application by integrating Google maps done successfully.

EX NO: 10

Mini Projects involving Flutter/Kotlin multi-platform

DATE:

JAVA CODE:

```
package com.example.db;
import android.app.Activity;
import android.app.AlertDialog.Builder;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends Activity implements OnClickListener {
    EditText Rollno, Name, Marks;
    Button Insert, Delete, Update, View, ViewAll;
    SQLiteDatabase db;

    /**
     * Called when the activity is first created.
     */
    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);
        Rollno = (EditText) findViewById(R.id.Rollno);
        Name = (EditText) findViewById(R.id.Name);
        Marks = (EditText) findViewById(R.id.Marks);
        Insert = (Button) findViewById(R.id.Insert);
        Delete = (Button) findViewById(R.id.Delete);
        Update = (Button) findViewById(R.id.Update);
        View = (Button) findViewById(R.id.View);
        ViewAll = (Button) findViewById(R.id.all);
        Insert.setOnClickListener(this);
        Delete.setOnClickListener(this);
        Update.setOnClickListener(this);
        View.setOnClickListener(this);
        ViewAll.setOnClickListener(this);
        // Creating database and table
        db = openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS student(rollno VARCHAR,name
        VARCHAR, marks VARCHAR);");
    }
}
```

```

public void onClick(View view) {
    // Inserting a record to the Student table
    if (view == Insert) {
        // Checking for empty fields
        if (Rollno.getText().toString().trim().length() == 0 ||
            Name.getText().toString().trim().length() == 0 ||
            Marks.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter all values");
            return;
        }
        db.execSQL("INSERT INTO studet VALUES('"+ Rollno.getText() +
            "'"+ Name.getText() + "','"+ Marks.getText() + "');");
        showMessage("Success", "Record added");
        clearText();
    }
    // Deleting a record from the Student table
    if (view == Delete) {
        // Checking for empty roll number
        if (Rollno.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter Rollno");
            return;
        }
        Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno='"+
            Rollno.getText() + "'", null);
        if (c.moveToFirst()) {
            db.execSQL("DELETE FROM student WHERE rollno='"+ Rollno.getText() +
                "'");
            showMessage("Success", "Record Deleted");
        } else {
            showMessage("Error", "Invalid Rollno");
        }
        clearText();
    }
    // Updating a record in the Student table
    if (view == Update) {
        // Checking for empty roll number
        if (Rollno.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter Rollno");
            return;
        }
        Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno='"+
            Rollno.getText() + "'", null);
        if (c.moveToFirst()) {
            db.execSQL("UPDATE student SET name='"+ Name.getText() + "','marks='"+
                Marks.getText() + "' WHERE rollno='"+ Rollno.getText() + "'");
        }
    }
}

```

```

        showMessage("Success", "Record Modified");
    } else {
        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}
// Display a record from the Student table
if (view == View)

    // Checking for empty roll number
    if (Rollno.getText().toString().trim().length() == 0) {
        showMessage("Error", "Please enter Rollno");
        return;
    }
    Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno=" +
Rollno.getText() + "'", null);
    if (c.moveToFirst()) {
        Name.setText(c.getString(1));
        Marks.setText(c.getString(2));
    } else {
        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}

if(view == ViewAll)
{
    c = db.rawQuery("SELECT * FROM student", null);
    if(c.getCount()==0)
    {
        showMessage("Error", "No records found");
        return;
    }
    StringBuffer buffer=new StringBuffer();
    while(c.moveToNext())
    {
        buffer.append("Rollno: "+c.getString(0)+"\n");
        buffer.append("Name: "+c.getString(1)+"\n");
        buffer.append("Marks: "+c.getString(2)+"\n\n");
    }
    showMessage("Student Details", buffer.toString());
}
}

public void showMessage(String title,String message)
{
    Builder builder=new Builder(this);
    builder.setCancelable(true);

```

```

        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
    public void clearText()
    {
        Rollno.setText("");
        Name.setText("");
        Marks.setText("");
        Rollno.requestFocus();
    }
}

```

XML CODE

```

<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="50dp"
        android:layout_y="20dp"
        android:text="Student Details"
        android:textSize="30sp" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="110dp"
        android:text="Enter Rollno:"
        android:textSize="20sp" />
    <EditText
        android:id="@+id/Rollno"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="100dp"
        android:inputType="number"
        android:textSize="20sp"
        android:text="rollno"

    />
    <TextView
        android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="160dp"
        android:text="Enter Name:"
        android:textSize="20sp" />
<EditText
    android:id="@+id/Name"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="150dp"
    android:inputType="text"
    android:text="Enter Name:"
    android:textSize="20sp" />
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="20dp"
    android:layout_y="210dp"
    android:text="Enter Marks:"
    android:textSize="20sp" />
<EditText
    android:id="@+id/Marks"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:text="Enter Mark:"
    android:layout_x="175dp"
    android:layout_y="200dp"
    android:inputType="number"
    android:textSize="20sp" />
<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:text="Insert"
    android:textSize="30dp" />
<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"
    android:text="Delete"
    android:textSize="30dp" />

```

```
<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:text="Update"
    android:textSize="30dp" />
```

```
<Button
    android:id="@+id/View"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="400dp"
    android:text="View"
    android:textSize="30dp" />
```

```
<Button
    android:id="@+id/all"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="500dp"
    android:text="all"
    android:textSize="30dp" />
```

```
</AbsoluteLayout>
```

OUTPUT:

6:40 AM 100%

Student Details

Enter Rollno: 1002

Enter Name: rani

Enter Marks: 99

INSERT DELETE

UPDATE VIEW

1 2 3 -

4 5 6 ↵

7 8 9 ✕

, 0 . →

6:39 AM 100%

Student Details

Enter Rollno:

Student Details

Rollno: 1001
Name: kani
Marks: 100

Rollno: 1002
Name: rani
Marks: 99

Rollno: 1003
Name: mani
Marks: 98

ALL

RESULT :

Thus the Mini Projects involving Flutter/Kotlin multi-platform completed successfully.